IT Project Management

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Lecture 02

Information Technology Context and Process Groups in Project Management

Learning Objectives

- Describe the systems view of project management and how it applies to information technology (IT) projects
- Understand organizations, including the four frames, organizational structures, and organizational culture
- Explain why stakeholder management and top management commitment are critical for a project's success

Learning Objectives

- Understand the concept of a project phase and the project life cycle, and distinguish between project development and product development
- Discuss the unique attributes and diverse nature of IT projects
- Describe recent trends affecting IT project management, including globalization, outsourcing, virtual teams, and agile project management

Project managers

Managing cross-functional teams effectively requires the project manager to:

Clarify goals

Get team members with the right skills

Measure progress

Recognize efforts

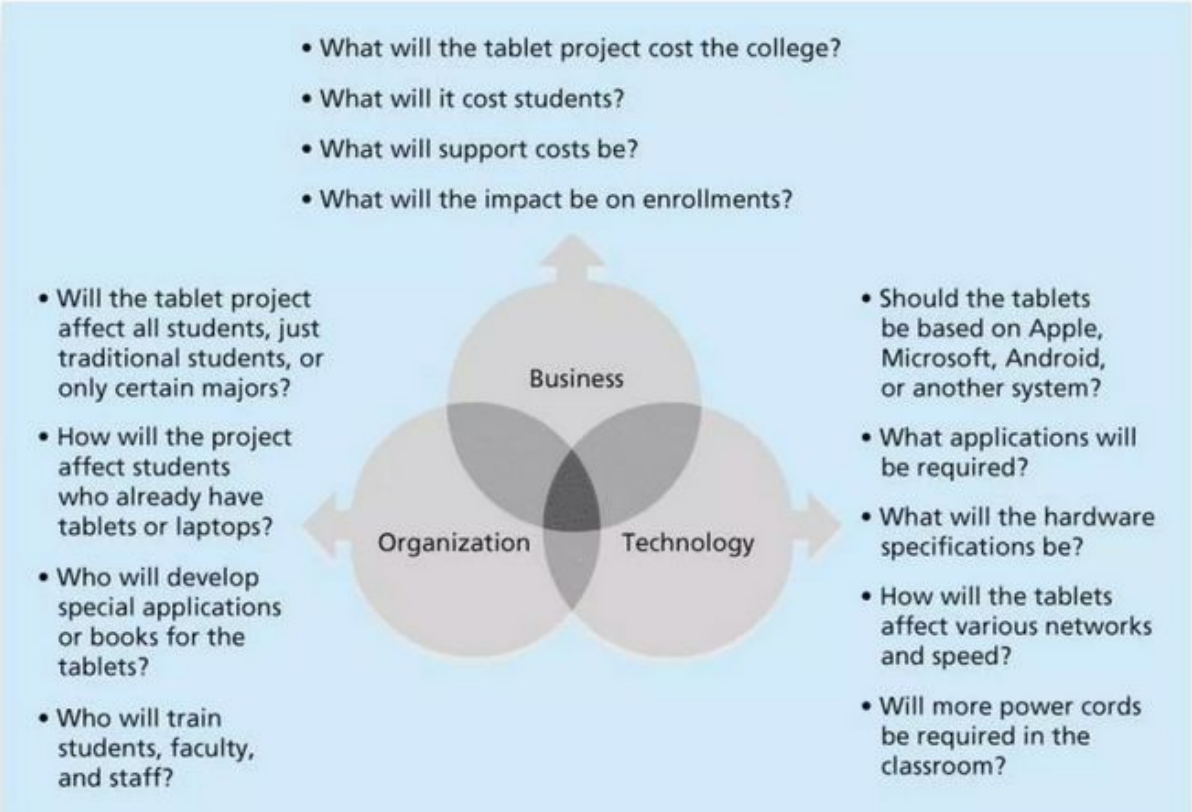
Projects Cannot Be Run In Isolation

- Projects must operate in a broad organizational environment
- Project managers need to use **systems thinking:**
  - taking a holistic view of carrying out projects within the context of the organization
- Senior managers must make sure projects continue to support current business needs

A Systems View of Project Management

- A **systems approach** emerged in the 1950s to describe a more analytical approach to management and problem solving
- Three parts include:
  - **Systems philosophy**: an overall model for thinking about things as systems
  - **Systems analysis**: problem-solving approach
  - **Systems management**: address business, technological, and organizational issues before making changes to systems

Figure 2-1. Three Sphere Model for Systems Management



- In a paper titled “A Study in Project Failure,” two researchers examined the success and failure of 214 IT projects over an eight-year period in several European countries.
- The researchers found that only one in eight (12.5 percent) were considered successful in terms of meeting scope, time, and cost goals.
- The authors said that the culture within many organizations is often to blame
- Among other things, people often do not discuss important leadership, stakeholder, and risk management issues

Organizational Structures

- 3 basic organization structures
  - **Functional:** functional managers report to the CEO
  - **Project:** program managers report to the CEO
  - **Matrix:** middle ground between functional and project structures; personnel often report to two or more bosses; structure can be weak, balanced, or strong matrix

Figure 2-3. Functional, Project, and Matrix Organizational Structures

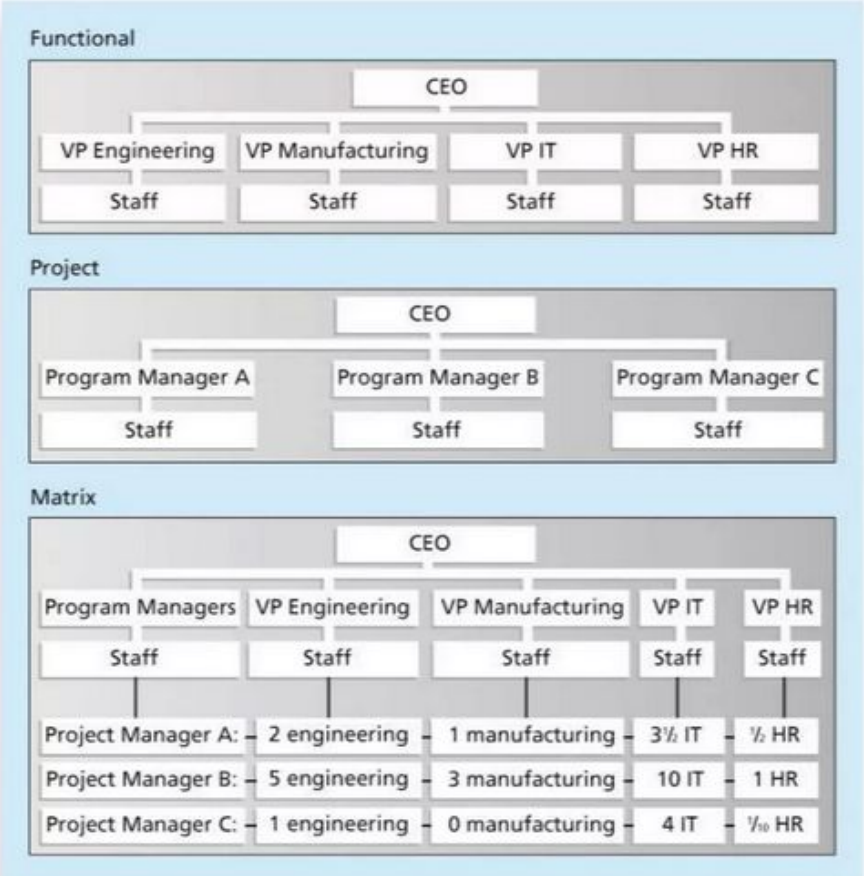


Table 2-1. Organizational Structure Influences on Projects

Project Characteristics	Organizational Structure Type				
	Functional	Matrix			Project
		Weak Matrix	Balanced Matrix	Strong Matrix	
Project manager's authority	Little or none	Limited	Low to moderate	Moderate to high	High to almost total
Percent of organization's personnel assigned full-time to project work	Virtually none	0–25%	15–60%	50–95%	85–100%
Who controls the project budget	Functional manager	Functional manager	Mixed	Project manager	Project manager
Project manager's role	Part-time	Part-time	Full-time	Full-time	Full-time
Common title for project manager's role	Project coordinator/ project leader	Project coordinator/ project leader	Project manager/ project officer	Project manager/ program manager	Project manager/ program manager
Project management administrative staff	Part-time	Part-time	Part-time	Full-time	Full-time

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Organizational Culture

- **Organizational culture** is a set of shared assumptions, values, and behaviors that characterize the functioning of an organization
- Many experts believe the underlying causes of many companies' problems are not the structure or staff, but the culture

Ten Characteristics of Organizational Culture

- Member identity*
- Group emphasis*
- People focus
- Unit integration*
- Control
- Risk tolerance*
- Reward criteria*
- Conflict tolerance*
- Means-ends orientation
- Open-systems focus*

*Project work is most successful in an organizational culture where these items are strong/high and other items are balanced.

Stakeholder Management

- Project managers must take time to identify, understand, and manage relationships with all project stakeholders
- Using the four frames of organizations can help meet stakeholder needs and expectations
- Senior executives/top management are very important stakeholders
- See Chapter 13, Project Stakeholder Management, for more information

Media Snapshot

- The media have often reported on mismanaged IT projects. A classic example and popular case study is the baggage handling system at Denver International Airport (DIA).
- The system was supposed to reduce flight delays, shorten waiting times at luggage carousels, and save money, but instead it caused huge problems.
- One important reason for this famous project disaster was the failure to recognize the project's complexity.

The Importance of Top Management Commitment

- People in top management positions are key stakeholders in projects
- A very important factor in helping project managers successfully lead projects is the level of commitment and support they receive from top management
- Without top management commitment, many projects will fail.
- Some projects have a senior manager called a **champion** who acts as a key proponent for a project.

How Top Management Can Help Project Managers

- Providing adequate resources
- Approving unique project needs in a timely manner
- Getting cooperation from other parts of the organization
- Mentoring and coaching on leadership issues

Best Practice

- **IT governance** addresses the authority and control for key IT activities in organizations, including IT infrastructure, IT use, and project management
- A lack of IT governance can be dangerous, as evidenced by three well-publicized IT project failures in Australia (Sydney Water's customer relationship management system, the Royal Melbourne Institute of Technology's academic management system, and One.Tel's billing system)

Need for Organizational Commitment to Information Technology (IT)

- If the organization has a negative attitude toward IT, it will be difficult for an IT project to succeed
- Having a Chief Information Officer (CIO) at a high level in the organization helps IT projects
- Assigning non-IT people to IT projects also encourage more commitment

Need for Organizational Standards

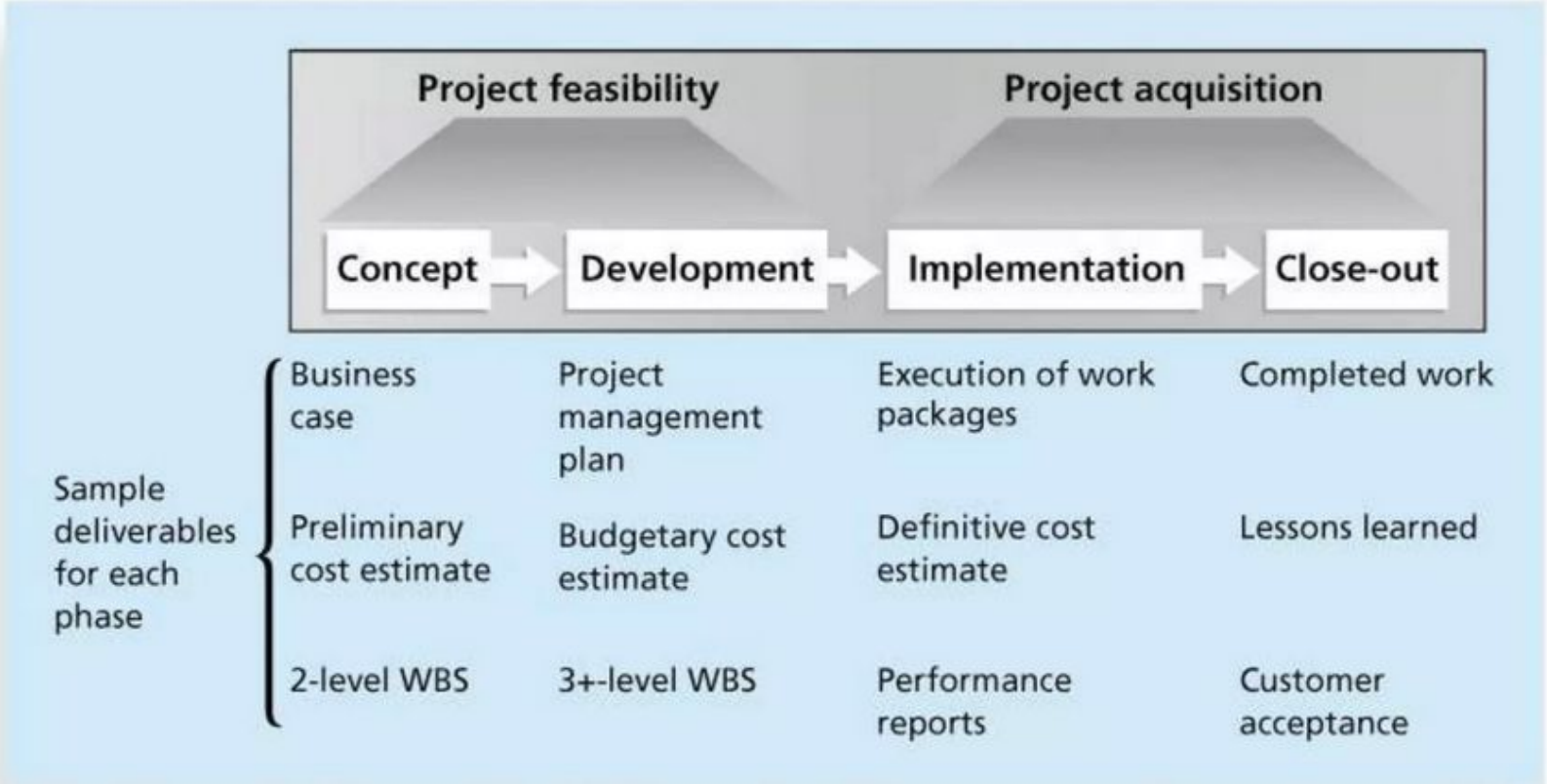
- Standards and guidelines help project managers be more effective
- Senior management can encourage
  - the use of standard forms and software for project management
  - the development and use of guidelines for writing project plans or providing status information
  - the creation of a project management office or center of excellence

Project Phases and the Project Life Cycle

- A **project life cycle** is a collection of project phases that defines
  - what work will be performed in each phase
  - what deliverables will be produced and when
  - who is involved in each phase, and
  - how management will control and approve work produced in each phase
- A **deliverable** is a product or service produced or provided as part of a project

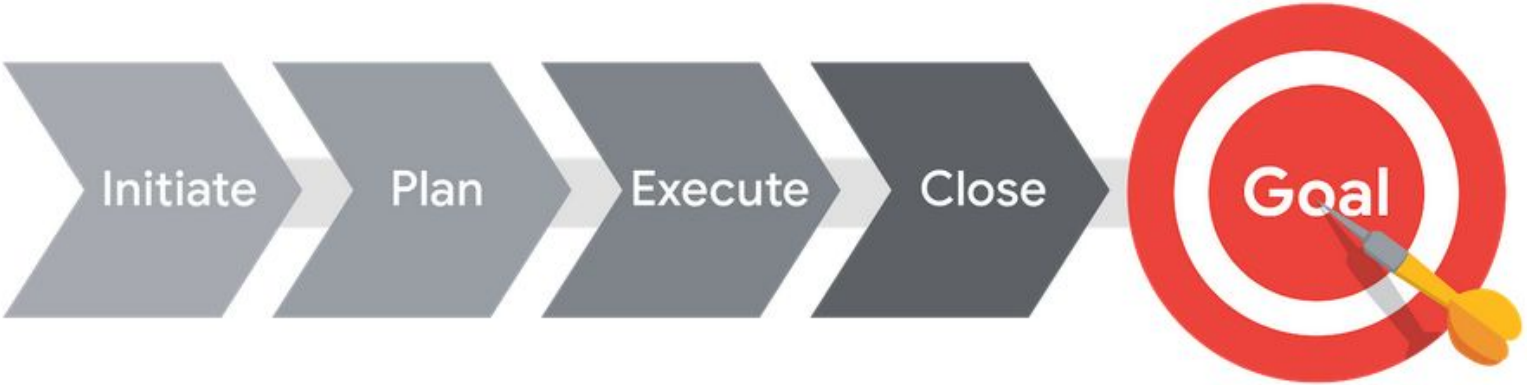
- In early phases of a project life cycle
  - resource needs are usually lowest
  - the level of uncertainty (risk) is highest
  - project stakeholders have the greatest opportunity to influence the project
- In middle phases of a project life cycle
  - the certainty of completing a project improves
  - more resources are needed
- The final phase of a project life cycle focuses on
  - ensuring that project requirements were met
  - the sponsor approves completion of the project

Figure 2-4. Phases of the Traditional Project Life Cycle



The **project life cycle** is the path for your project from start to finish. Each project phase builds toward the subsequent phase and helps to create a structure for the project. To recap, the main phases of the project life cycle are: initiating the project, making a plan, executing and completing tasks, and closing the project.

In this reading, we will summarize each phase of the project life cycle.



The project life cycle

Initiate the project

In this phase, ask questions to help set the foundation for the project, such as:

- Who are the stakeholders?
- What are the client's or customer's goals?
- What is the purpose and mission of the project?
- What are the measurable objectives for the team?
- What is the project trying to improve?
- When does this project need to be completed?
- What skills and resources will the project require?
- What will the project cost? What are the benefits?

Make a plan

In this phase, make a plan to get your project from start to finish.

- Create a detailed project plan. What are the major milestones? What tasks or deliverables make up each milestone?
- Build out the schedule so you can properly manage the resources, budget, materials, and timeline. Here, you will create an itemized budget.

Execute the project

In this phase, put all of your hard work from the first two phases into action.

- Monitor your project team as they complete project tasks.
- Break down any barriers that would slow or stop the team from completing tasks.
- Help keep the team aware of schedule and deliverable expectations.
- Address weaknesses in your process or examine places where your team may need additional training to meet the project's goals.
- Adapt to changes in the project as they arise.

Close the project

In this phase, close out the project.

- Identify that your team has completed all of the requested outcomes.
- Release your team so they can support other projects within the company.
- Take time with your team to celebrate your successes!
- Pass off all remaining deliverables and get stakeholder approval.
- Document the lessons you and your team learned during the project.
- Reflect on ways to improve in the future.

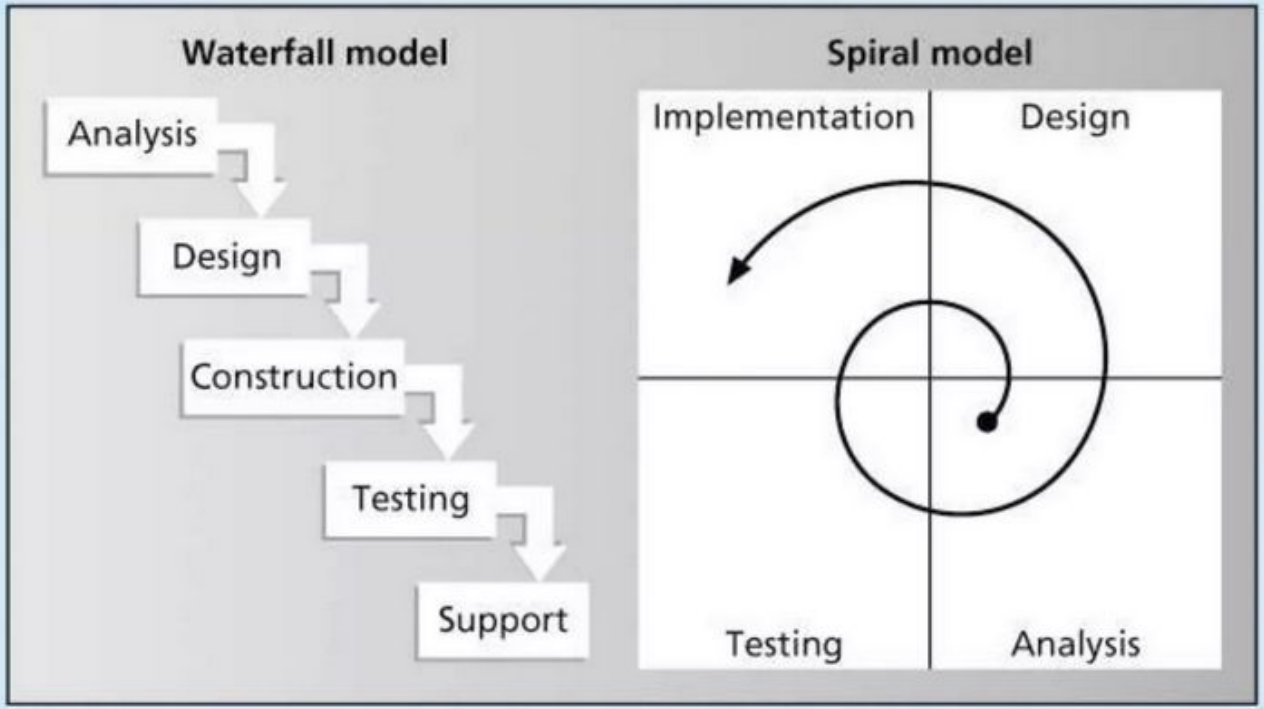
Product Life Cycles

- Products also have life cycles
- The **Systems Development Life Cycle (SDLC)** is a framework for describing the phases involved in developing and maintaining information systems
- Systems development projects can follow
  - **Predictive life cycle**: the scope of the project can be clearly articulated and the schedule and cost can be predicted
  - **Adaptive Software Development (ASD) life cycle**: requirements cannot be clearly expressed, projects are mission driven and component based, using time-based cycles to meet target dates

Predictive Life Cycle Models

- Waterfall model: has well-defined, linear stages of systems development and support
- Spiral model: shows that software is developed using an iterative or spiral approach rather than a linear approach
- Incremental build model: provides for progressive development of operational software
- Prototyping model: used for developing prototypes to clarify user requirements
- Rapid Application Development (RAD) model: used to produce systems quickly without sacrificing quality

Figure 2-5. Waterfall and Spiral Life Cycle Models



Agile Software Development

- Agile software development has become popular to describe new approaches that focus on close collaboration between programming teams and business experts
- See the last section of this chapter and Chapter 3 for more information on agile

The Importance of Project Phases and Management Reviews

- A project should successfully pass through each of the project phases in order to continue on to the next
- Management reviews, also called **phase exits** or **kill points**, should occur after each phase to evaluate the project's progress, likely success, and continued compatibility with organizational goals

The Context of IT Projects

- IT projects can be very diverse in terms of size, complexity, products produced, application area, and resource requirements
- IT project team members often have diverse backgrounds and skill sets
- IT projects use diverse technologies that change rapidly. Even within one technology area, people must be highly specialized

Recent Trends Affecting IT Project Management

- Globalization
- Outsourcing: **Outsourcing** is when an organization acquires goods and/or sources from an outside source. **Offshoring** is sometimes used to describe outsourcing from another country
- Virtual teams: A **virtual te**am is a group of individuals who work across time and space using communication technologies
- Agile project management

Important Issues and Suggestions Related to Globalization

- Issues
  - Communications
  - Trust
  - Common work practices
  - Tools
- Suggestions
  - Employ greater project discipline
  - Think global but act local
  - Keep project momentum going
  - Use newer tools and technology

Global Issues

- Outsourcing also has disadvantages. For example, Apple benefits from manufacturing products in China, but it had big problems there after its iPhone 4S launch in January 2012 caused fighting between migrant workers who were hired by scalpers to stand in line to buy the phones.
- When Apple said it would not open its store in Beijing, riots resulted and people attacked security guards. The Beijing Apple Store has had problems before. In May 2011, four people were injured when a crowd waiting to buy the iPad 2 turned ugly.

Virtual Teams Advantages

- Increasing competiveness and responsiveness by having a team of workers available 24/7
- Lowering costs because many virtual workers do not require office space or support beyond their home offices.
- Providing more expertise and flexibility by having team members from across the globe working any time of day or night
- Increasing the work/life balance for team members by eliminating fixed office hours and the need to travel to work.

Virtual Team Disadvantages

- Isolating team members
- Increasing the potential for communications problems
- Reducing the ability for team members to network and transfer information informally
- Increasing the dependence on technology to accomplish work
- See text for a list of factors that help virtual teams succeed, including team processes, trust/relationships, leadership style, and team member selection

Agile Project Management

- Agile means being able to move quickly and easily, but some people feel that project management, as they have seen it used, does not allow people to work quickly or easily.
- Early software development projects often used a waterfall approach, as defined earlier in this chapter. As technology and businesses became more complex, the approach was often difficult to use because requirements were unknown or continuously changing.
- Agile today means using a method based on iterative and incremental development, in which requirements and solutions evolve through collaboration.

Agile Makes Sense for Some Projects, But Not All

- Many seasoned experts in project management warn people not to fall for the hype associated with Agile.
- For example, J. Leroy Ward, Executive Vice President at ESI International, said that "Agile will be seen for what it is … and isn't.…Project management organizations embracing Agile software and product development approaches will continue to grow while being faced with the challenge of demonstrating ROI through Agile adoption."*

*J. Leroy Ward, "The Top Ten Project Management Trends for 2011," projecttimes.com (January 24, 2011).

Manifesto for Agile Software Development

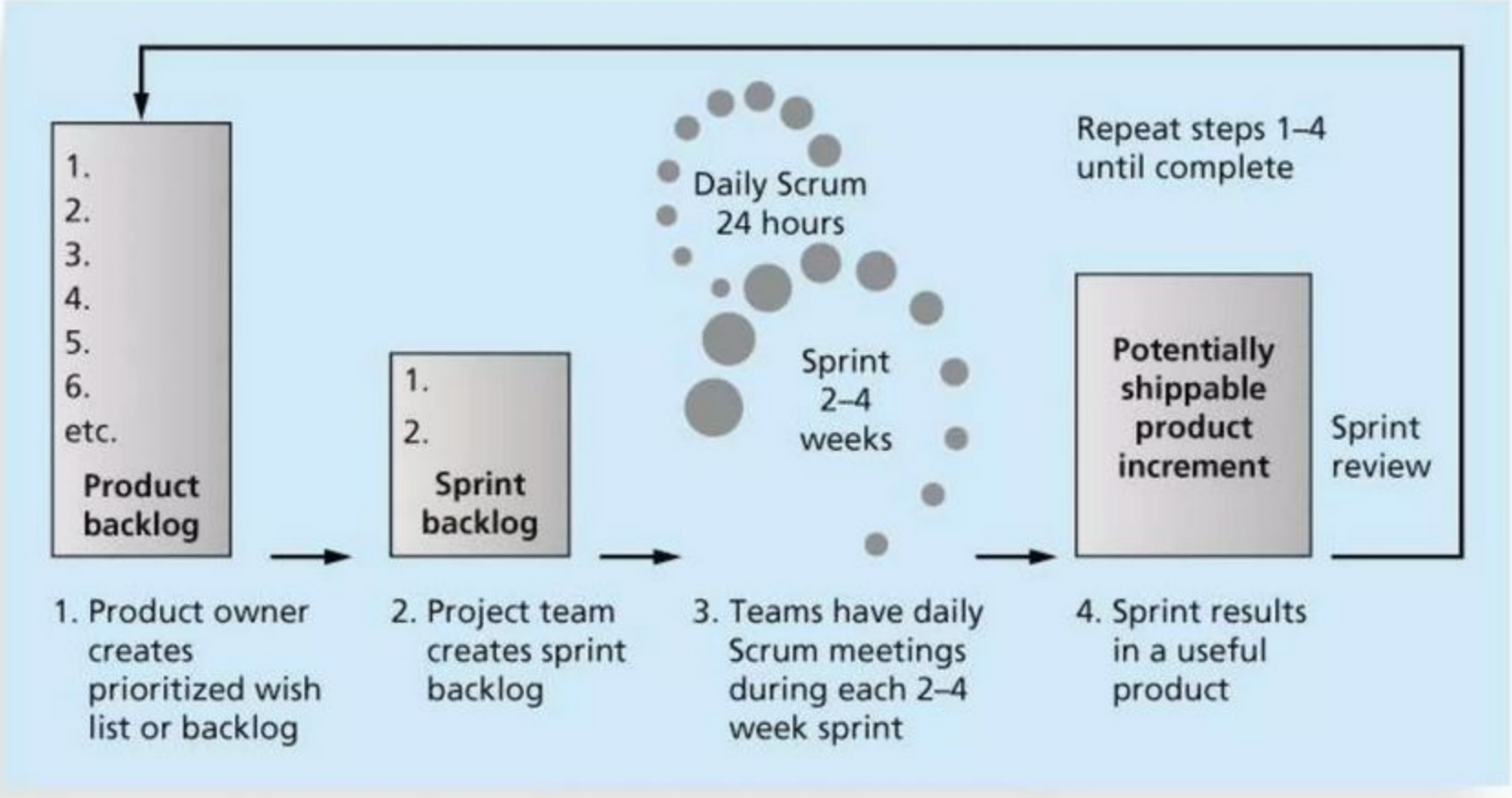
- In February 2001, a group of 17 people that called itself the Agile Alliance developed and agreed on the Manifesto for Agile Software Development, as follows:
- “We are uncovering better ways of developing software by doing it and helping others do it. Through this work we have come to value:
- Individuals and interactions over processes and tools
- Working software over comprehensive documentation
- Customer collaboration over contract negotiation
- Responding to change over following a plan”*

*Agile Manifesto, www.agilemanifesto.org

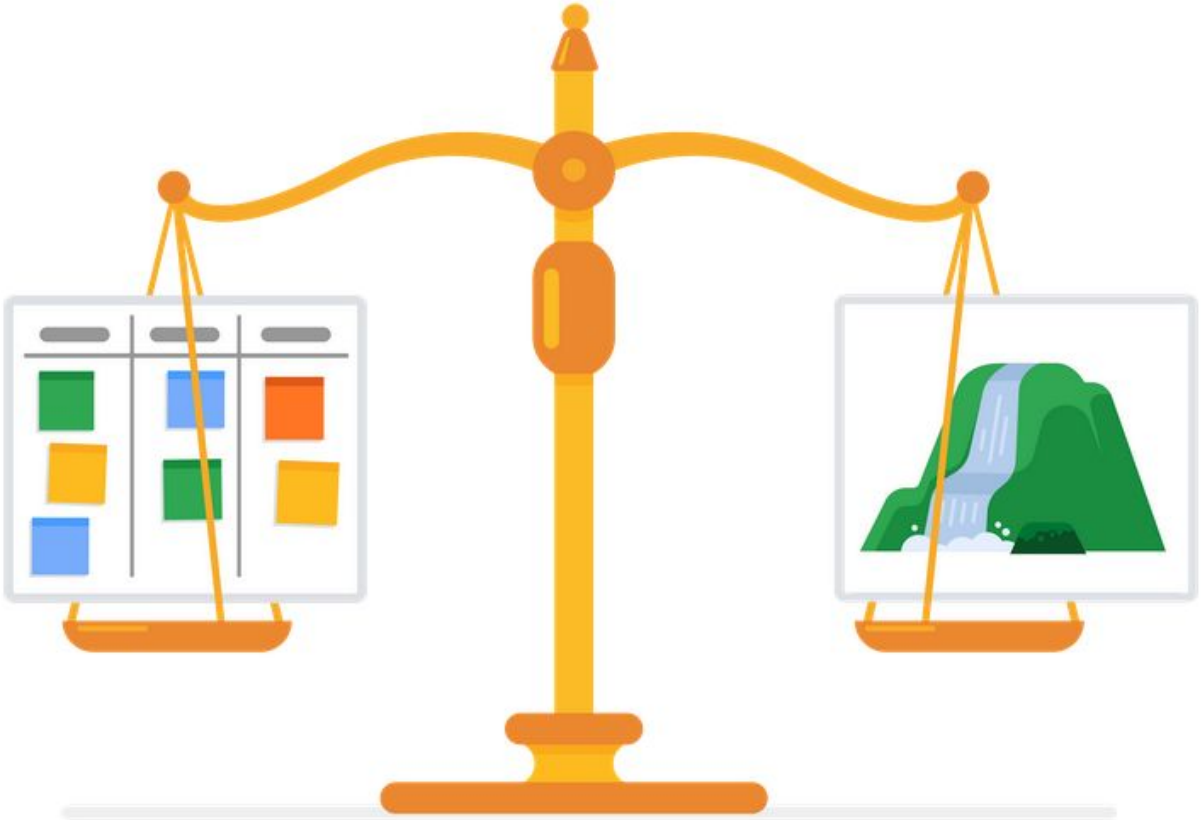
Scrum

- According to the Scrum Alliance, Scrum is the leading agile development method for completing projects with a complex, innovative scope of work.
- The term was coined in 1986 in a Harvard Business Review study that compared high-performing, cross-functional teams to the scrum formation used by rugby teams.

Figure 2-6. Scrum Framework



Now that you know more about some of the different approaches and frameworks associated with project management, let's compare specific aspects of **Waterfall** (also commonly called **traditional**) and **Agile** approaches.



	Waterfall	Agile
Project manager's role	Project manager serves as an active leader by prioritizing and assigning tasks to team members.	Scrum Master acts primarily as a facilitator, removing any barriers the team faces. Team shares more responsibility in managing their own work.
Scope	Project deliverables and plans are well-established and documented in the early stages of initiating and planning. Changes go through a formal change request process.	Planning happens in shorter iterations and focuses on delivering value quickly. Subsequent iterations are adjusted in response to feedback or unforeseen issues.
Schedule	Follows a mostly linear path through the initiating, planning, executing, and closing phases of the project.	Time is organized into phases called Sprints. Each Sprint has a defined duration, with a set list of deliverables planned at the start of the Sprint.
Cost	Costs are kept under control by careful estimation up front and close monitoring throughout the life cycle of the project.	Costs and schedule could change with each iteration.
Quality	Project manager makes plans and clearly defines criteria to measure quality at the beginning of the project.	Team solicits ongoing stakeholder input and user feedback by testing products in the field and regularly implementing improvements.

Communication	Project manager continually communicates progress toward milestones and other key indicators to stakeholders, ensuring that the project is on track to meet the customer's expectations.	Team is customer-focused, with consistent communication between users and the project team.
Stakeholders	Project manager continually manages and monitors stakeholder engagement to ensure the project is on track.	Team frequently provides deliverables to stakeholders throughout the project. Progress toward milestones is dependent upon stakeholder feedback.

Lean and Six Sigma methodologies

Previously you learned about Agile and Waterfall project management approaches. Now, we will define some key concepts from Lean and Six Sigma methodologies. We will learn how these methodologies can be used to organize and manage your projects, and we will discuss which is the most effective for different kinds of projects.

Lean

Lean methodology is often referred to as Lean Manufacturing because it originated in the manufacturing world. The main principle in Lean methodology is the removal of waste within an operation. By optimizing process steps and eliminating waste, only value is added at each phase of production.

Today, the Lean Manufacturing methodology recognizes eight types of waste within an operation: defects, excess processing, overproduction, waiting, inventory, transportation, motion, and non-utilized talent. In the manufacturing industry, these types of waste are often attributed to issues such as:

- Lack of proper documentation
- Lack of process standards
- Not understanding the customers' needs
- Lack of effective communication
- Lack of process control
- Inefficient process design
- Failures of management

These same issues create waste in project management.

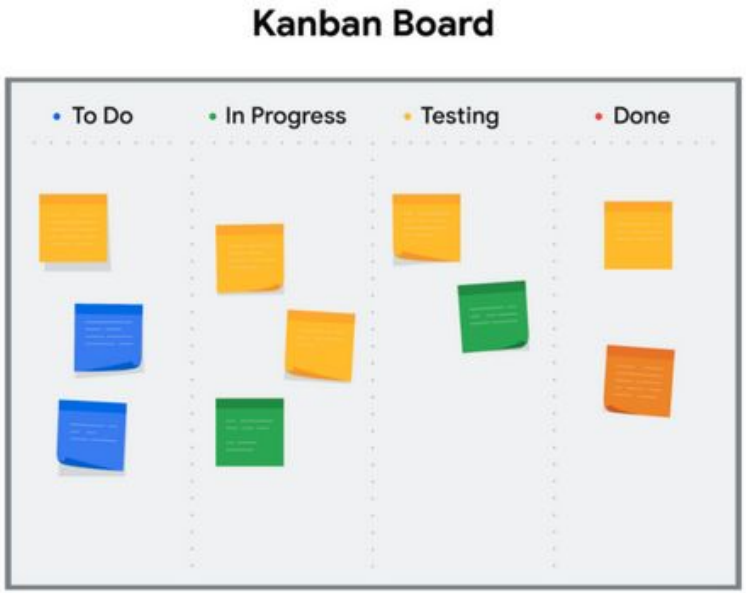
Implement Lean project management when you want to use limited resources, reduce waste, and streamline processes to gain maximum benefits.

You can achieve this by using the pillars of the Lean 5S quality tool. The term 5S refers to the five pillars that are required for good housekeeping: sort, set in order, shine, standardize, and sustain. Implementing the 5S method means cleaning up and organizing the workplace to achieve the smallest amount of wasted time and material. The 5S method includes these five steps:

1. **Sort:** Remove all items not needed for current production operations and leave only the bare essentials.
2. **Set in order:** Arrange needed items so that they are easy to use. Label items so that anyone can find them or put them away.
3. **Shine:** Keep everything in the correct place. Clean your workspace every day.
4. **Standardize:** Perform the process in the same way every time.
5. **Sustain:** Make a habit of maintaining correct procedures and instill this discipline in your team.

Within the Lean methodology, 5S helps you boost performance.

The final concept of Lean uses a **Kanban** scheduling system to manage production. The Kanban scheduling system, or Kanban board, is a visualization tool that enables you to optimize the flow of your team's work. It gives the team a visual display to identify what needs to be done and when. The Kanban board uses cards that are moved from left to right to show progress and help your team coordinate the work.



Six Sigma

Six Sigma is a methodology used to reduce variations by ensuring that quality processes are followed every time. The term “Six Sigma” originates from statistics and generally means that items or processes should have 99.9996% quality.

The seven key principles of Six Sigma are:

1. Always focus on the customer.
2. Identify and understand how the work gets done. Understand how work really happens.
3. Make your processes flow smoothly.
4. Reduce waste and concentrate on value.
5. Stop defects by removing variation.
6. Involve and collaborate with your team.
7. Approach improvement activity in a systematic way.

Use this methodology to find aspects of the product or process that are *measurable* like time, cost, or quantity. Then inspect that measurable item and reject any products that do not meet the Six Sigma standard. Any process that created unacceptable products has to be improved upon.

Lean Six Sigma

After both Lean and Six Sigma were put into practice, it was discovered that the two methodologies could be combined to increase benefits. The tools used in Lean, such as Kanban boards and 5S, build quality in processes from the beginning. Products developed using Lean methods are then inspected or tested using Six Sigma standards. The products that do not meet these standards are rejected.

The largest difference between these methodologies is that Lean streamlines processes while Six Sigma reduces variation in products by building in quality from the beginning and inspecting products to ensure quality standards are met. You may find that one of these two methods—or using them both together—can improve the efficiency of your projects.